

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE –
RAIGAD -402 103
Summer Supplimentary –2023

Branch: Electronics and Telecommunication Engineering **Sem.:- VII**

Subject with Subject Code: - Digital Communication BTETC701 **Marks: 60**

Date:-13/07/2023

Time:- 3 Hr.

Instructions to the Students

1. Each question carries 12 marks.
2. Attempt **ALL** questions.
3. Illustrate your answers with neat sketches, diagram etc., wherever necessary.
4. If some part or parameter is noticed to be missing, you may appropriately assume it and should mention it clearly

- | | (Level) | (Marks) |
|---|---------|---------|
| Q.1 Solve Any Two Questions from the following: | | |
| A) A Television signal with a bandwidth of 4.2 MHz is transmitted using binary PCM. The number of quantization levels is 512. Calculate:
1. Code word length
2. Transmission bandwidth
3. Final bit rate
4. Output signal to quantization noise ratio | L2 | (06) |
| B) Explain the basic digital communication system with neat block diagram. | L1 | (06) |
| C) Differentiate between Analog and Digital Communication. | L1 | (06) |
| Q.2 Solve Any Two Questions from the following: | | |
| A) Explain T- carrier system. | L2 | (06) |
| B) Explain Jamming. What are the various types of Jammers. | L3 | (06) |
| C) Explain the term: PSD. State and prove the properties of PSD. | L2 | (06) |
| Q.3 Solve Any Two Questions from the following: | | |
| A) Explain the working of QPSK receiver with neat block diagram. Also explain the carrier synchronization in QPSK. | L3 | (06) |

- B) A voice signal band limited to 3.4Khz is sampled at 8 kHz and pulse code modulated using 64 quantization levels. Ten such signals are time division multiplexed using one 5-bit synchronizing word. Calculate minimum channel bandwidth. L3 (06)
- C) Explain the term Narrow-band noise. How is it expressed in terms of quadrature components? L2 (06)

Q.4 Solve **Any Two Questions** from the following:

- A) Explain the term: CDF. State and prove the properties of CDF. L1 (06)
- B) Explain PCM system. L1 (06)
- C) Explain the working of BFSK receiver with neat block diagram. L2 (06)

Q.5 Solve **Any Two Questions** from the following:

- A) Explain sampling theorem with neat diagrams in time and frequency domain. L2 (06)
- B) Explain the working of Additive Scramblers. L1 (06)
- C) Explain the working of DS-SS Scheme with neat block diagram. L2 (06)
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